COMSATS University Islamabad Vehari

Department of Computer Science

Assignment 1 (LAB) – Spring 2026

Subject: Programming Fundamentals (CSC-103) **Instructor: Dr. Muhammad Mudassar**

Class BSCS Batch SP26 **Marks=10**

Course Learning Outcomes (CLOs): CLO-4 "Implement a program using programing constructs."

Instructions:

1. Create an integrated PDF of programs, outputs, and answers.
2. Use CamScanner or similar software for PDF creation.
3. It is mandatory to upload the PDF to the CUOnline portal.
4. To avoid portal problems, submit your task before the deadline.
5. The class name of every program should by your complete name as camel notation
6. Use proper names for the variables (this will have marks)
7. Deadline: 27 March 2026 (Friday) till 04:00 PM.

Question 1:

Write a Java program that calculates the sum of all odd numbers from 1 to a number entered by the user. Complete the following requirements:

- Prompt the user for an integer input.
- Use a for loop to iterate from 1 up to the entered number.
- Summing only the odd numbers.
- Use break statement if the sum increases more than 500.
- Use continue statement to skip all those odd numbers ending in 7.

After the loop completes, display the final sum.

Question 2: You are building a program that calculates the Train ticket price based on the customer's age and Distance he wants to travel. The program should prompt the user to enter their age and then ask their destination in Kilometers and then determine the ticket price based on the following rules for age:

- If the age is below 12, the ticket price is 20 RS per Kilometer.
- If the age is between 12 and 60, the ticket price is 50 RS per Kilometer.
- If the age is above 60, the ticket price is 10 RS per Kilometer.

Use switch statement.

Question 3: Zakat is calculated after a specific gold/silver/amount a user have after one year. Write a java program to calculate Zakat for a user based on following criteria**. (user switch to get the following 4 options and inside the switch user of-else to calculate the zakat). Show the final amount to be paid as zakat to the user.**

1. **User have only gold.**

Ask the user how many Tola of gold he have? If user have 7.5 Tola of gold, he has to pay 2.5% of its total amount as zakat. (amount of 1 Tola = 4,35,000)

2. **User have only silver**

Ask the user how many Tola of Silver he have? If user have 52 Tola of silver, he has to pay 2.5% of its total amount as zakat. (amount of 1 Tola = 5500 RS)

3. **User have cash only**

Ask the user he have how much cash? If user have 1,80,000 cash, he has to pay 2.5% of his amount as zakat.

4. **User have some cash and have any amount of gold or silver**

Ask the user about how much cash, gold and silver he belongs. Calculate the total price of gold and silver and them sum all these amounts along with cash to get the **total assets**. Calculate the 2.5% of total assets as zakat to be paid.

Question 4: Write a java program to implement a number guessing game, where the computer assumes an integer number using the Random function and user is prompted to guess the number by entering a number in the provided range.

- Computer generates a random number (1-100)
- Player has 5 attempts to guess it

The flow of the game will be:

1. Welcome message & instructions
2. Generate random number
3. Prompt for guess → Get user input
4. Provide feedback (too high/too low)
5. Repeat until win or attempts exhausted
6. Ask "Play again?" → Update gameActive

Question 5: Write a program that prompts the user to enter a three-digit integer and determines whether it is a palindrome number. A number is palindrome if it reads the same from right to left and from left to right. Here is a sample run of this program:

```
Enter a three-digit integer: 121
121 is a palindrome
```

```
Enter a three-digit integer: 123
123 is not a palindrome
```

Question 6: (Count positive and negative numbers and compute the average of numbers) Write a program that reads an unspecified number of integers, determines how many positive and negative values have been read, and computes the total and average of the input values (not counting zeros). Your program ends with the input 0. Display the average as a floating-point number. Here is a sample run:

```
Enter an integer, the input ends if it is 0: 1 2 -1 3 0 Enter
The number of positives is 3
The number of negatives is 1
The total is 5.0
The average is 1.25
```

```
Enter an integer, the input ends if it is 0: 0 Enter
No numbers are entered except 0
```

Question 7:

(*Find the highest score*) Write a program that prompts the user to enter the number of students and each student's name and score, and finally displays the name of the student with the highest score.

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